

Clinical NLP - Summarization

Reading doctors' reports and creating short, easy summaries with NLP

Dataset: Medical_Reportss.csv • 3,176 clinical follow-up reports across 10 conditions

Introduction

Medical reports are written in **plain, everyday language** by doctors — with symptoms, diagnosis, treatment, and follow-up notes. Computers can't understand plain text directly, so we use NLP to turn these reports into a structured form the model can actually work with.

It turns messy, unorganized reports into useful, structured information.

It saves doctors time and helps them make quicker, better decisions.

Why Is This Needed?

Hospitals generate thousands of medical reports daily.

Reading long reports takes a lot of time.

Doctors need quick access to key information.

AI can automatically summarize reports and even predict the medical condition.

The Problem

Medical reports are usually long and unstructured.

Important clinical information is difficult to identify quickly.

Manual analysis is time-consuming.

So we need an automated system that can:

- Predict diseases

- Extract key findings

- Generate summaries

Objective

Build a Clinical NLP application using Python.

Preprocess medical reports.

Predict medical conditions using Machine Learning.

Extract important clinical findings.

Generate AI-based summaries.

Create an interactive Streamlit web application.

Tools & Libraries

NLTK (Natural Language Toolkit) – a Python library for text processing. Here, its sentence tokenizer splits each report into sentences so we can pull out the key ones. NLTK also covers other common NLP techniques:

- Tokenization – splits a sentence into individual words (tokens)

- Stopwords – removes common filler words like “the”, “is”, “a”

- Stemming – trims a word down to its root, e.g. running → run, treatment → treat

- Lemmatization – converts a word to its dictionary form, e.g. headaches → headache

Hugging Face Transformers – ready-made AI models such as DistilBART and T5, used to generate short summaries

TF-IDF (Term Frequency–Inverse Document Frequency) – turns text into numbers (up to 10,000 word combinations), giving more weight to important words and automatically skipping common filler words

Plus scikit-learn for the machine learning model, and Python, NumPy, and Pandas for handling the data

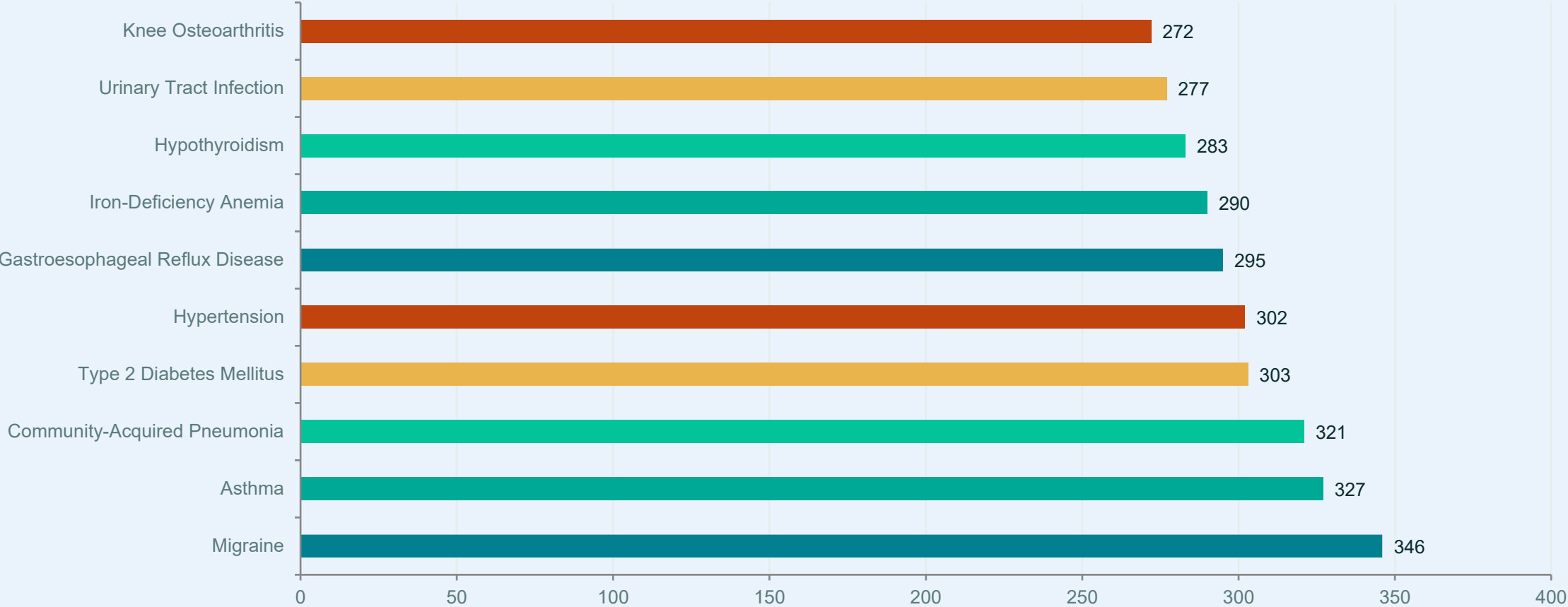
Dataset

Medical reports were used as the input dataset.

Each report contains patient symptoms, diagnosis, treatment, and follow-up information.

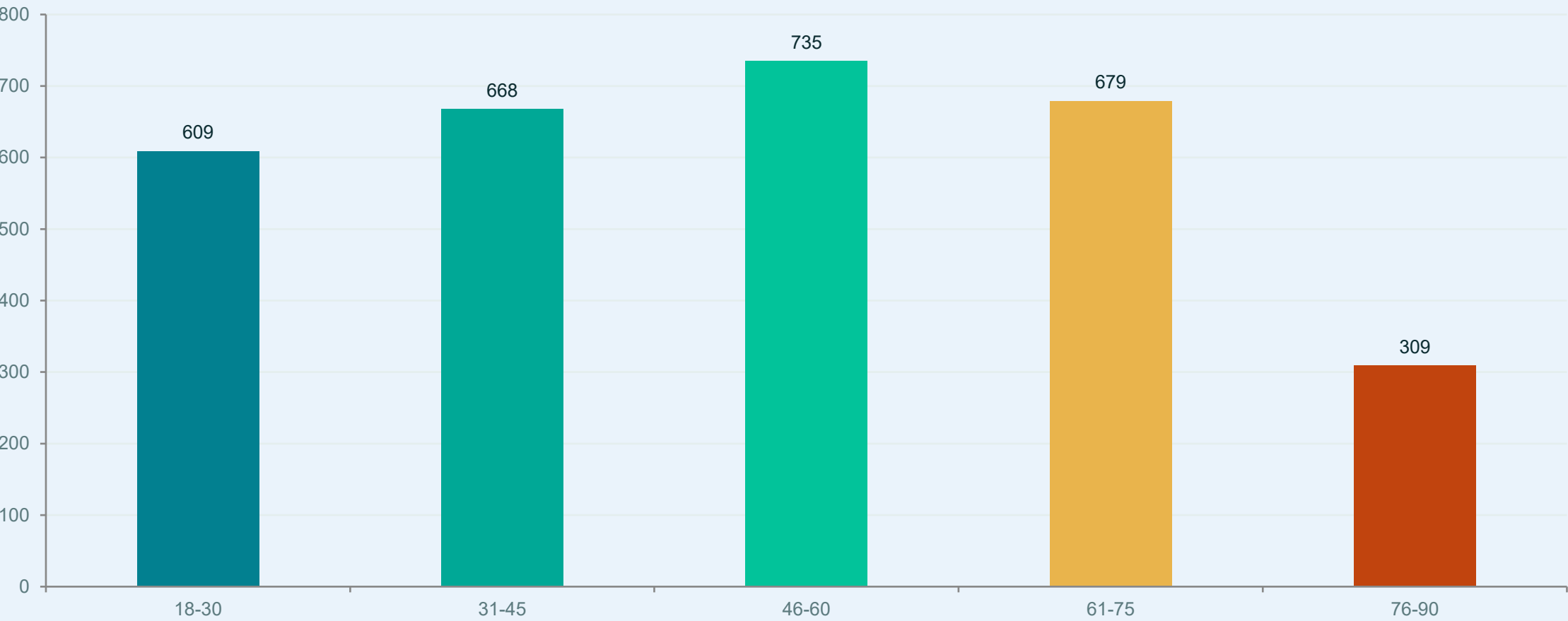
Disease labels are used to train the prediction model.

Reports per Condition



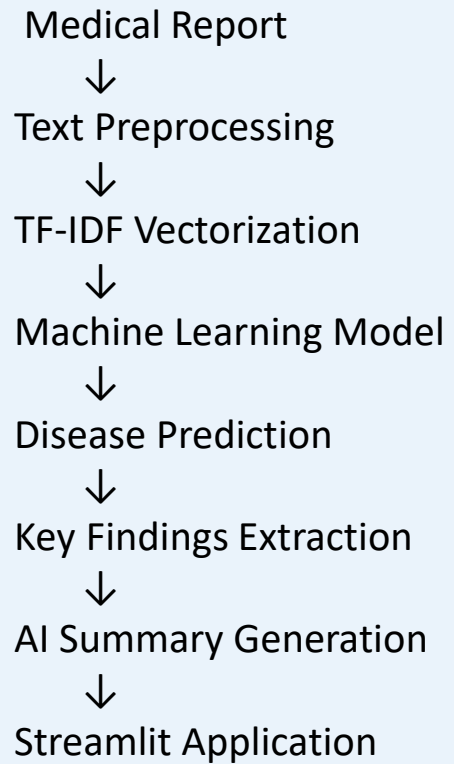
Migraine and Asthma are the most represented conditions; distribution is reasonably balanced across all 10 categories.

Age Distribution



Patient age ranges 18–82 years (mean $\approx$ 49.7); most patients fall in the 46–60 bracket.

Work Flow



Data Cleaning

Remove duplicate reports

Drop rows with missing report text

Fill missing Condition / Gender with “Unknown”

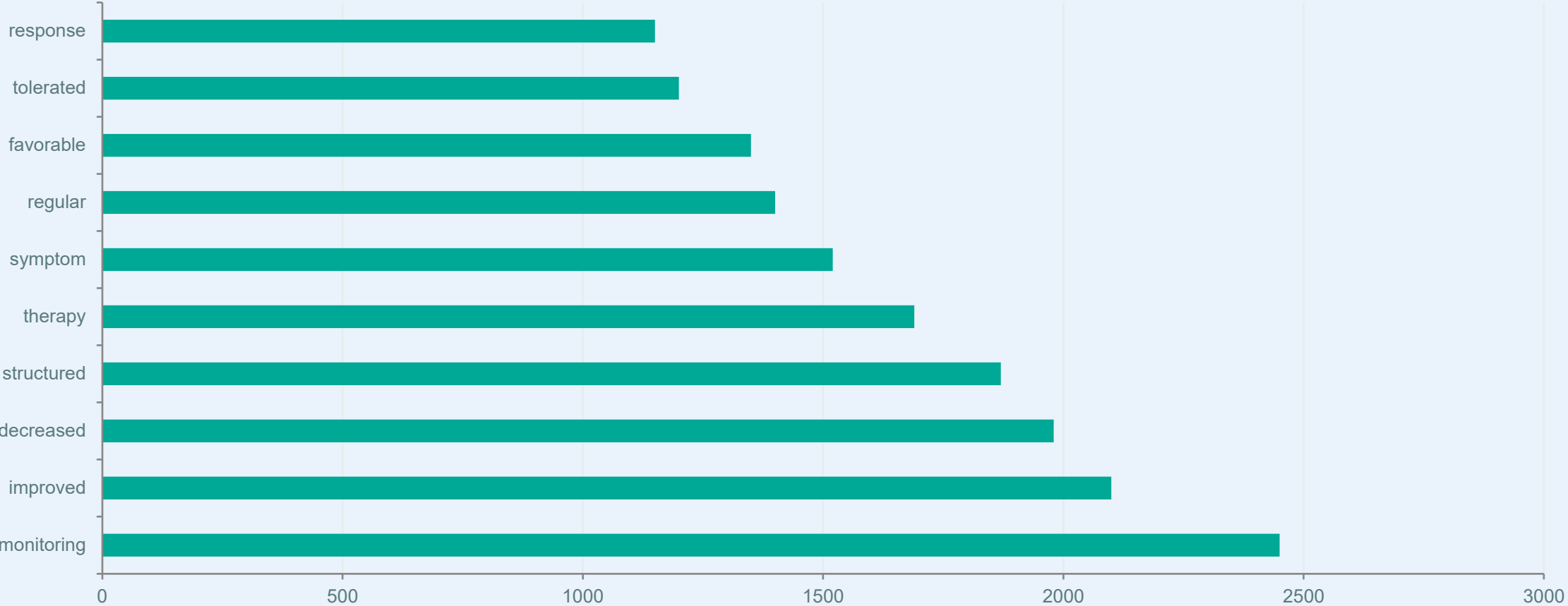
Fill missing Age / Follow-up weeks with the median value

Collapse extra spaces in the text

Remove unwanted symbols, keeping useful punctuation

TF-IDF then filters out common filler words automatically

Most Frequent Clinical Terms



After removing common stopwords, words like “improved,” “decreased,” and “tolerated” show up most — strong clues about how patients respond to treatment.

Feature Extraction

Machine Learning models cannot understand text directly.

TF-IDF converts text into numerical vectors (up to 10,000 features, using 1-2 word combinations).

Gives higher weight to important words.

Reduces the importance of common words.

Disease Prediction

Reports are split: 80% for training, 20% for testing.

A machine learning model predicts the disease from the text.

TF-IDF vectors are used as input, and the model learns patterns from labeled medical reports.

We use Logistic Regression — a simple, fast algorithm that works great with **TF-IDF**. It trains quickly, even on **large datasets**, and gives good accuracy for predicting medical conditions.

It's also less likely to overfit than more complex models, and it's easy to interpret — which is why it's such a popular starting point in NLP.

Key Clinical Extraction and Summary of Reports

Each report is split into sentences. Any sentence with a clinical keyword is kept, such as:

“symptom”, “pain”

“severity”, “tolerated”

“diagnosed”, “assessment”

“treatment”, “response”

“follow-up”, “improved”, “decreased”

Example — Migraine: nausea, headache, reduced activity → symptom diary, sleep counseling, preventive medication

This saves doctors time and highlights the key clinical details.

Long reports are then summarized using an AI model.

We use a Transformer model (DistilBART) to generate a short, meaningful summary of each report.

Results of Streamlit

 **Clinical NLP**

This application uses Machine Learning and Natural Language Processing to analyze medical reports.

🌟 **Features**

✂ Medical Text Preprocessing

🩺 Condition Prediction

🔑 Key Findings Extraction

📄 AI-Generated Summary

📶 **Model Status**

✅ Condition Model: Ready

✅ AI Summarizer: Ready

Deploy

Automated Medical Report Summarization and Key Findings Extraction

 **Medical Report Analysis**

Choose an example medical report:

Migraine Report

Edit the report or paste your own medical report:

The patient presented with persistent migraine headaches occurring three to four times per week. Pain severity was rated 8/10. The patient also reported nausea and sensitivity to light.

The patient was diagnosed with chronic migraine. Treatment was initiated with prescribed medication.

During follow-up, headache frequency decreased significantly and the patient reported improvement in symptoms. The treatment was well tolerated and no serious adverse effects were reported.

📄 Words

66

🔤 Characters

478

📄 Sentences

7

Analyze Medical Report

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41

 Analyze Medical Report

 Medical report analysis completed!

Predicted Medical Condition

Migraine

Key Clinical Findings

1. The patient presented with persistent migraine headaches occurring three to four times per week.
2. Pain severity was rated 8/10.
3. The patient also reported nausea and sensitivity to light.
4. The patient was diagnosed with chronic migraine.
5. Treatment was initiated with prescribed medication.

AI-Generated Medical Summary

A patient at the Royal Victoria Hospital in Belfast has been treated for a severe migraine, according to the NHS.

▼  View Original Medical Report

The patient presented with persistent migraine headaches occurring three to four times per week. Pain severity was rated 8/10. The patient also reported nausea and sensitivity to light.

The patient was diagnosed with chronic migraine. Treatment was initiated with prescribed medication.

During follow-up, headache frequency decreased significantly and the patient reported improvement in symptoms. The treatment was well tolerated and no serious adverse effects were reported.

>  View Preprocessed Medical Report

Clinical NLP: Automated Medical Report Summarization

▼ View Preprocessed Medical Report

The patient presented with persistent migraine headaches occurring three to four times per week. Pain severity was rated 8/10. The patient also reported nausea and sensitivity to light. The patient was diagnosed with chronic migraine. Treatment was initiated with prescribed medication. During follow-up, headache frequency decreased significantly and the patient reported improvement in symptoms. The treatment was well tolerated and no serious adverse effects were reported.

Clinical NLP: Automated Medical Report Summarization

Conclusion

- Built a working Clinical NLP application using **Python** and **Streamlit**
- The model predicts medical conditions with an **accuracy of 94.8%** — a strong result.
- This can help doctors save time and read medical reports more efficiently.